HARISANKAR HARIKUMAR

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Nationality: Indian | German: B1 (Goethe-Zertifikat) | English: C1

PROFESSIONAL PROFILE

Master's student in Remote Sensing and Geoinformatics at Karlsruhe Institute of Technology (KIT), specializing in Computer Vision and Environmental Geodesy. Experienced in Earth Observation, with a focus on SAR and optical satellite data, multisensor data fusion, and AI-based image analysis. Strong practical background in Python and PyTorch, with experience developing automated EO processing and ML pipelines, deep-learning models, and systematic evaluation workflows. Completed two research internships at the German Aerospace Center (DLR) in land-cover classification, change detection, geospatial data preparation, and foundation-model-based image analysis, including a publication in the ISPRS Archives/Annals. Currently working at Fraunhofer IOSB on AI-based image analysis of hyperspectral and RGB data.

WORK EXPERIENCE

Student Assistant: AI based image processing for Hyperspectral and RGB Images

Fraunhofer Institute of Optronics, System Technologies and Image Exploitation(IOSB), Karlsruhe / *Aug 2026*

- Development of deep learning-based classification and segmentation models for hyperspectral and RGB sensor data using Python and PyTorch
- Processing and analysis of multi-modal imaging data from spectral cameras for automated material identification and quality inspection
- Implementation of data preprocessing, augmentation, and evaluation workflows for automated sensor-based classification and inspection systems

Internship: Vision Language Models for Satellite Imagery & Geodata Analysis

Remote Sensing Technology Institute (IMF), German Aerospace Center (DLR), Munich / *Jul - Nov 2025*

- Developed end-to-end Python pipelines for automated feature extraction, high-resolution imagery
- Fine-tuned Google PaliGemma2, a generative AI vision-language foundation model, for operational remote sensing tasks including prompt-based detection, segmentation, and automated scene description.
- Built scalable HPC pipelines for large satellite datasets, automating tiling, quality control, and multimodal label generation using Qwen-VL to create a 16,500-instance training dataset.
- Validated model outputs against reference data and documented performance metrics; research published in the ISPRS Archives/Annals

Internship: Remote Sensing Data Analysis for Land Cover Monitoring

German Remote Sensing Data Center (DFD), German Aerospace Center (DLR), Munich / *Oct – Dec 2024*

- Processed and analyzed multi-temporal multispectral satellite imagery for land cover classification and change detection using machine learning methods
- Implemented automated geodata processing and analysis workflows in Python for large-scale satellite dataset preprocessing
- Integrated OpenStreetMap and in-situ reference data with satellite datasets for training data preparation and validation of classification models
- Evaluated model transferability across temporal periods and documented classification accuracy metrics for systematic performance benchmarking

PUBLICATION

H. Harikumar, S. Bittner, et al. (2025). *From Pixels to Semantics: Can a Single Instruction-Tuned VLM Unify Geospatial Building Analysis Tasks.* ISPRS Archives/Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences. Publication link: H.Harikumar, K. Bittner, et al (2025)

ACADEMIC PROJECTS

- **Satellite-Derived Bathymetry and Coastal Water Quality Estimation from Multi-Temporal Sentinel-2 Imagery using Machine Learning**. A Comparative Study of Spectral Ratio, Random Forest, and Neural Network Approaches over the Mediterranean. GitHub link: sentinel2-water-quality-bathymetry
- **Multi-Sensor Data Fusion and Change Detection using Statistical (CUSUM) and Deep Learning (LSTM)** approaches by fusing radar and optical sensor time series for automated anomaly detection and method comparison (Python, Pandas, PyTorch). GitHub link: Deforestation-monitoring-Sentinel-1and-2-fusion
- **Forest Structure Assessment using SAR and LiDAR Tomography** - vertical canopy profiling and biomass-related parameter retrieval (Python, SNAP) . GitHub link: Lidar-and-SAR-Tomography
- **Persistent Scatterer InSAR for Surface Deformation Monitoring** - displacement time series analysis from SAR data (SNAP, StaMPS)
- **Land Cover Classification of Hyperspectral and Multispectral** Imagery - supervised classification and accuracy assessment (ImageJ)
- **Semantic Segmentation of Multispectral Satellite Imagery using CNNs** - pixel-wise land cover classification with deep learning (Python, PyTorch)
- **Peatland Condition Mapping using Sentinel-1 interferometric coherence and Sentinel-2 spectral time series** fused for wet/drained peatland classification with Random Forest, including carbon emission estimation (Python, GEE, SNAP, Scikit-learn, Rasterio) - **Ongoing**

EDUCATION

Master of Science in Remote Sensing and Geoinformatics

Karlsruhe Institute of Technology (KIT), Germany | Jun 2023 – Present / *Specialization: Computer Vision and Environmental Geodesy*

Bachelor of Engineering in Civil Engineering

JNN Institute of Engineering, Anna University, Chennai, India | Aug 2016 - Sep 2020 / *Specialization: Environmental Engineering & Surveying*

Additional Training: Practical Remote Sensing & Geometric Correction using ERDAS IMAGINE (BRS Labs / Hexagon) | Advanced GIS: ArcGIS, ArcPy & QGIS (NCTT, India) | Geo Informatics and Analytics (JNN) | Global Environmental Management (DTU / Coursera)

TECHNICAL SKILLS

Programming & ML: Python, PyTorch, NumPy, Pandas, Matplotlib, Scikit-learn, HPC/containerized workflows

Geospatial Libraries: Rasterio, GDAL, GeoPandas, Shapely, Fiona

Sensor Data and Analysis: ArcGIS, Multi-sensor data fusion, time series analysis, change detection, anomaly detection, classification, performance benchmarking, domain transferability evaluation, QGIS, SNAP, SAR processing, multispectral & hyperspectral analysis, land cover classification

AI & Computer Vision: Deep Learning, Transformer, generative AI, Foundation Model fine-tuning, semantic segmentation, object detection, domain adaptation, synthetic-to-real transfer, model evaluation

DevOps & Tools: Linux, Git, Docker, pytest, CUDA, HPC clusters (GPU acceleration), LaTeX

LANGUAGES

English (C1 - Professional) | **German** (B1 - Goethe-Zertifikat) | **Malayalam** (Native)